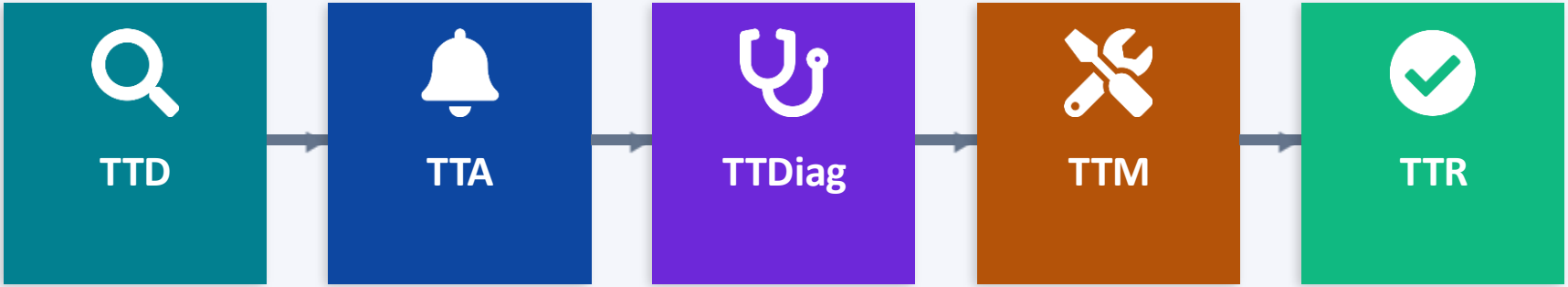
# KPIs of Incident Management

A Comprehensive Framework for Measuring & Improving Incident Response Performance

# OBJECTIVE

To establish a structured, measurable, and continuously improving incident management capability that minimizes customer impact through rapid detection, swift response, accurate diagnosis, effective mitigation, and thorough resolution, aligned with elite SRE practices.




*Each KPI builds on the previous — delays cascade downstream*

# AGENDA

## 01
### TTD – Time to Detect
*Observability & monitoring effectiveness*

## 02
### TTA – Time to Acknowledge
*On-call responsiveness & alert quality*

## 03
### TTDiag – Time to Diagnose
*Root cause identification & engineering maturity*

## 04
### TTM – Time to Mitigate
*Operational agility & service restoration*

## 05
### TTR – Time to Resolve
*End-to-end incident lifecycle performance*

## 06
### KPI Interdependencies & Bottlenecks
*Cascade effects, bottleneck patterns, optimization*

# 01 | TTD – Time to Detect

## Definition & Key Points

**Incident Occurrence → First Detection**
Measures observability effectiveness, not human performance.

**Why it matters:**

- Defines how long customers are impacted before detection
- Tied to monitoring coverage and signal quality
- Largest hidden contributor to downtime in alert-driven environments

**Common Pitfalls:**

- Measuring from alert firing vs. actual degradation
- Missing silent failures (no alert generated)
- Alert suppression masking detection delays

## Improvement Strategies

- Shift to symptom-based alerting (golden signals)
- Implement synthetic monitoring for user journeys
- Event correlation / AIOps for anomaly detection
- Distributed tracing for hidden failures

🎯 **Target: ≤ 5 minutes**

# 02 | TTA – Time to Acknowledge

## Definition & Key Points

**Detection → Human Acknowledgment**
Measures responsiveness of the support model.

**Why it matters:**
- Reflects on-call effectiveness
- Delays here stall all downstream recovery steps

**Key Insight:**
*Reducing alert noise improves TTA more than adding more people.*

**Pitfalls:**
- Auto-acknowledgment inflating performance
- Alerts routed to wrong teams

## Improvement Strategies

- Enforce high signal-to-noise alert policies
- Use tiered escalation with ownership mapping
- Track and eliminate flaky/duplicate alerts
- 24/7 on-call coverage with reliable escalation

🎯 **Target: ≤ 5 minutes**

# 03 | TTDiag – Time to Diagnose

## Definition & Key Points

**Acknowledgment → Root Cause Identification**
Measures system understandability & engineering maturity.

**Why it matters:**
- Typically the longest cognitive phase
- Strong predictor of both TTM and TTR

**Influencing Factors:**
- Observability depth (logs, traces, metrics correlation)
- System complexity (microservices vs monolith)
- Quality of runbooks & knowledge sharing

## Improvement Strategies

- Invest in end-to-end observability tooling
- Build & maintain diagnostic runbooks
- Blameless postmortems to improve future diagnosis
- Service dependency maps + automated root cause hints

🎯 **Target: ≤ 1 hour**

# 04 | TTM – Time to Mitigate

## Definition & Key Points

**Diagnosis → Service Restored via Workaround**
Measures operational agility, not perfection.

**Core Principle:**
*Mitigate first, fix later. Customers care about service restoration, not root cause.*

**Influencing Factors:**
- System resilience (failover, redundancy)
- Deployment rollback capabilities
- Decision-making authority during incidents
- Risk tolerance of the organization

## Improvement Strategies

- Predefine mitigation playbooks for known failures
- Enable feature flags, traffic shifting & rollbacks
- Run chaos engineering exercises regularly
- Reduce approval bottlenecks during active incidents

🎯 **Target: ≤ 3 hours**

# 05 | TTR – Time to Resolve

## Definition & Key Points

**Detection → Full Resolution**
Measures end-to-end incident lifecycle performance.

**Why it matters:**
- Represents total business impact duration
- Common executive-level KPI

**Key Dependencies:**
- All prior KPIs (TTD, TTA, TTDiag, TTM) feed into TTR
- Change management speed
- Fix complexity and release pipeline efficiency

## Improvement Strategies

- Optimize upstream KPIs first — biggest leverage
- Parallelize mitigation & root cause fix workflows
- Improve CI/CD release pipelines for fast fixes
- Validate true stability before declaring resolution

🎯 **Target: ≤ 6 hours**

# 06 | KPI Interdependencies & Bottleneck Patterns

**TTD** Detect → **TTA** Acknowledge → **TTDiag** Diagnose → **TTM** Mitigate → **TTR** Resolve

*A delay early in the pipeline multiplies downstream impact*

### 01 Invisible Incidents

Symptom: High TTD; everything else appears fine

**Root: Poor monitoring coverage**

### 02 Alert Chaos

Symptom: Good TTD, poor TTA

**Root: Alert fatigue & noise**

### 03 Diagnosis Paralysis

Symptom: Strong detection, weak TTDiag

**Root: Complexity + lack of observability**

### 04 Slow Recovery

Symptom: Fast diagnosis, slow TTM

**Root: Lack of operational readiness**

# Organizational & Cultural Differentiators

## HIGH-PERFORMING TEAMS

**Characteristics:**

- Blameless culture & psychological safety
- Strong on-call ownership model
- Continuous improvement mindset
- Heavy investment in observability

**Behaviors:**

- Treat alerts as product quality signals
- Regularly prune and tune alerts
- Conduct rigorous blameless postmortems
- "You build it, you run it" ownership model

## LOW-PERFORMING TEAMS

- Alert overload & reactive firefighting
- Knowledge silos between teams
- Slow decision-making during incidents

## Critical Cultural Factors

- Psychological Safety → faster diagnosis & escalation
- Ownership Model → improves TTA & TTDiag
- Learning Loops → postmortems feed monitoring (TTD) and runbooks (TTDiag & TTM)

# KPI Target Summary & Maturity Assessment

| KPI | Full Name | Your Target |
|---|---|---|
| TTD | Time to Detect | ≤ 5 min |
| TTA | Time to Acknowledge | ≤ 5 min |
| TTDiag | Time to Diagnose | ≤ 1 hr |
| TTM | Time to Mitigate | ≤ 3 hrs |
| TTR | Time to Resolve | ≤ 6 hrs |

**Overall: Your targets define a high-maturity, SRE-aligned incident management capability**

# KEY TAKEAWAYS

🔍 Detection is the foundation. TTD drives the entire incident timeline. Invest in observability first.

🔔 Alert quality > Alert quantity. Reducing noise improves TTA more than adding headcount.

🧠 Diagnosis is the longest phase. End-to-end observability and runbooks are force multipliers for TTDiag.

⚡ Mitigate first, fix later. Customer restoration (TTM) takes priority over root cause perfection.

🏆 The system is the strategy. Elite TTR comes from optimizing the full pipeline, not any single KPI.